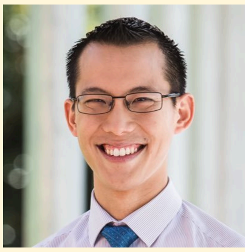
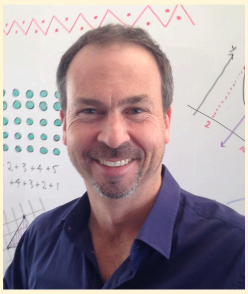
Maths Week Australia

Ambassador Problem Picks



Eddie Woo shares The Two Rectangles problem

Kaye Stacey shares a Paper Folding Challenge



James Tanton shares the Finding Paths problem

Michaela Epstein shares the Spirolaterals challenge

Eddie Woo's Favourite Problem: The Two Rectangles

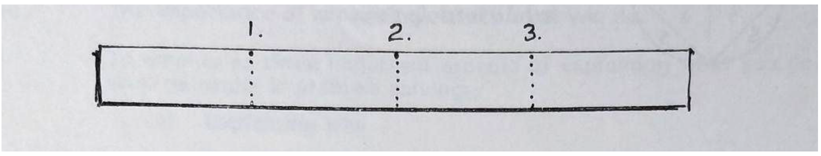
Create two rectangles so that the first has a bigger perimeter, and the second has a bigger area.

Some questions to get you thinking:

- Compare your rectangles with other students'. What do you notice?
- Find some more rectangle pairs. Can you find a shortcut?
- Can you create two rectangles so that the first has exactly double the perimeter of the second, and the second has exactly double the area of the first?

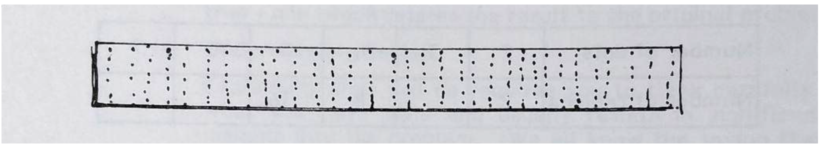
Kaye Stacey's Favourite Problem: The Paper Folding Challenge

Take a strip of paper about 30 cm long and 2 cm wide. Fold it in half, then half again. When you unfold it you will see that 3 creases have been made.



SOURCE: *Strategies for Problem Solving* (Stacey and Groves, 2006). Reproduced with permission of the author.

If you folded the strip in half **10 times**, how many creases would there be?



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Some questions to get you thinking:

- How many folds is it possible to actually make?
- How might you keep track of what you're doing?
- Can you see a pattern? This can help with working systematically.
- What if you wanted to fold the strip in half 25 times? 100 times? *N* times?

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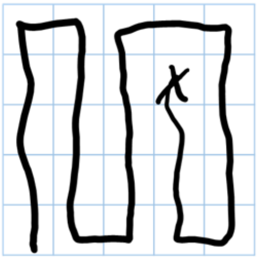
James Tanton's Favourite Problem: Finding Paths

Pick a spot (any spot!) on a 5x5 grid . Can you walk a path (moving left, right, up or down) that follows each and every cell exactly once?

Here's a path that works:



Some questions to get you thinking:

- Are all starting points possible? (Are there any shortcuts you can take to work this out?)
- Are there any starting points where it's impossible to find a path?
- What if you didn't have a 5x5 grid, but had a...
    - 6x6 grid?
    - 5x6 grid?
    - 5x5x5 cube, where you visit each smaller cube?

[Learn more from James about this problem here.]

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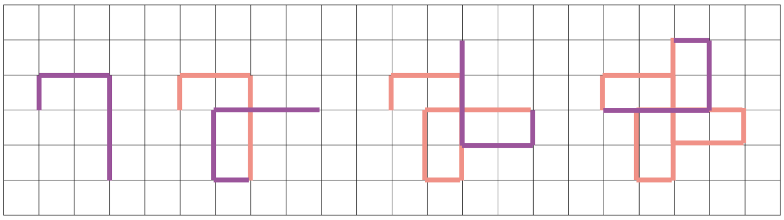
Michaela Epstein's Favourite Problem: Spirolaterals

Use three numbers to take a walk around a grid, and something interesting emerges.



This is a (1, 2, 3) Spirolateral. It's made by moving:

1. One space up,
2. Turning 90° clockwise. Moving 2 spaces to the right
3. Turning 90 ° clockwise. Moving 3 spaces down
4. Repeating steps 1-3 (turning 90 ° and moving 1, 2 or 3 spaces each time).

Pick any three numbers to make your own Spirolaterals. What do you notice? What shapes can you make?

Some questions to get you thinking:

- Can you make a spirolateral that looks like a square?
- Can you predict what shape you will make based on the 3 numbers you choose?
- What happens if you make a Spirolateral with 4 numbers? Or 5? 6? ...?
- What changes if you rotate by 60° instead of 90°? Or another angle?

[Learn more from Michaela about this problem here.]